

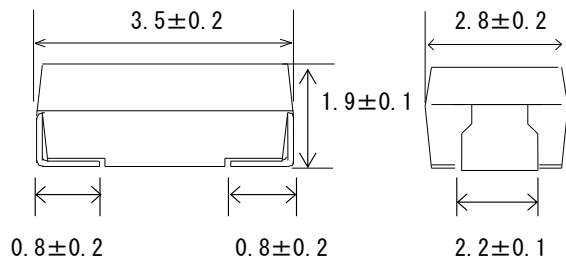
NeoCapacitor® Conductive Chip Polymer Tantalum Capacitor PSH series

PSHB21E156M090B [B2 -case 15 μ F / 25 V]

Halogen-free, Lead-free, and RoHS Compliant

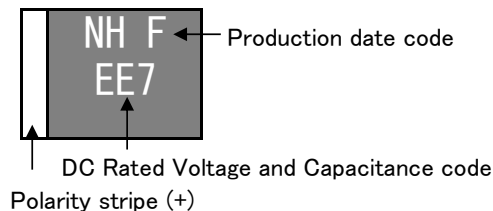
Feb-25

Out view design (Dimensions : mm)

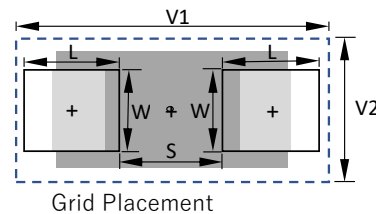


Note) Aforementioned dimension is recommended value.
(JEITA RC-2371A is also recommended for reference.)

Marking



Recommended Pattern design (mm)



	W	L	S	V1	V2
Maximum (Most) Land	2.35	2.21	0.92	6.32	4.00
Median (Nominal) Land	2.23	1.80	1.12	5.22	3.50
Minimum (Least) Land	2.13	1.42	1.28	4.36	3.24

Production date code

	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
2023	a	b	c	d	e	f	g	h	j	k	l	m
2024	n	p	q	r	s	t	u	v	w	x	y	z
2025	A	B	C	D	E	F	G	H	J	K	L	M
2026	N	P	Q	R	S	T	U	V	W	C	Y	Z

NOTE: Production date code will resume for beginning in 2027.

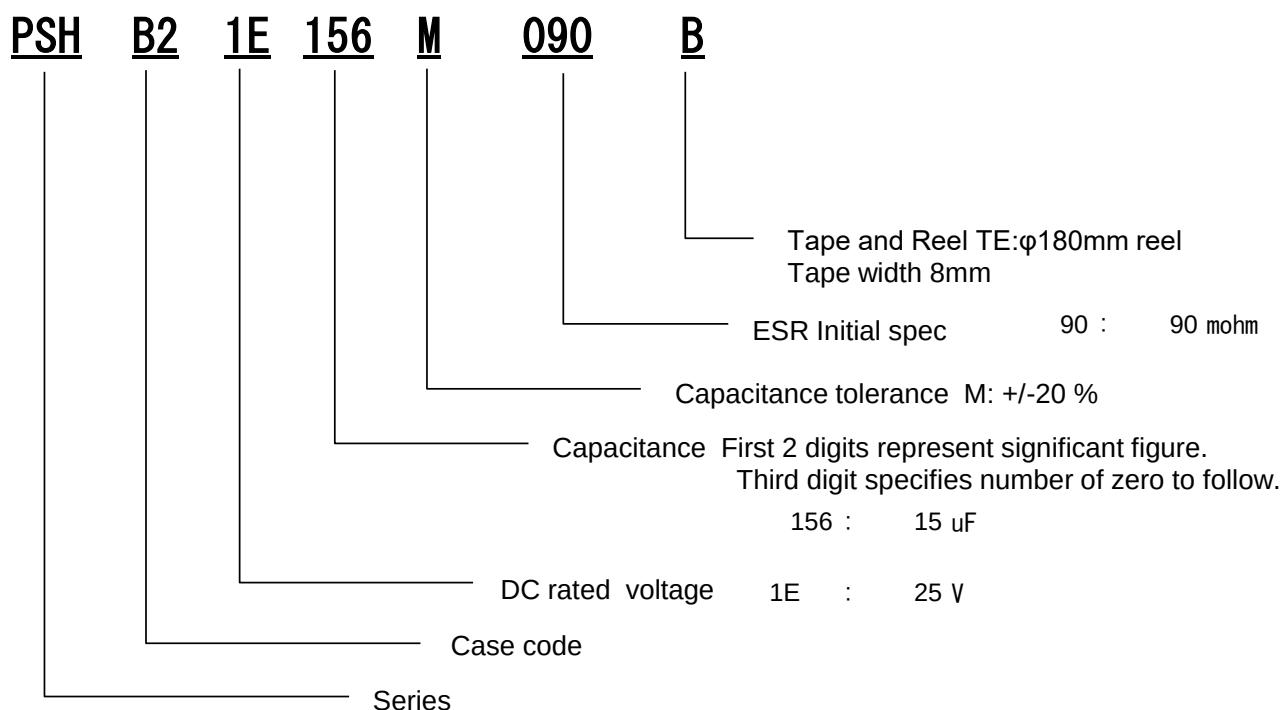
Standard ratings

Rated Voltage	Capacitance* ¹ (μ F, 120Hz)	Leakage Current* ² (μ A, +25deg.C)	Dissipation Factor ($\tan\delta$)* ² (120Hz)			ESR* ² (mohm, +25deg.C, 100kHz)	Permissible Ripple Current (mA Arms Max 25deg.C, 100kHz)	$\Delta C/C$ (1)
			25degC	-55degC	105degC			
25	15	112.5	0.1	0.1	0.15	90	972	±20%

*1 : Tolerance ±20%, *2 : Initial characteristic
 $\Delta C/C(1)$ at Shock, at Surge Voltage, at Endurance

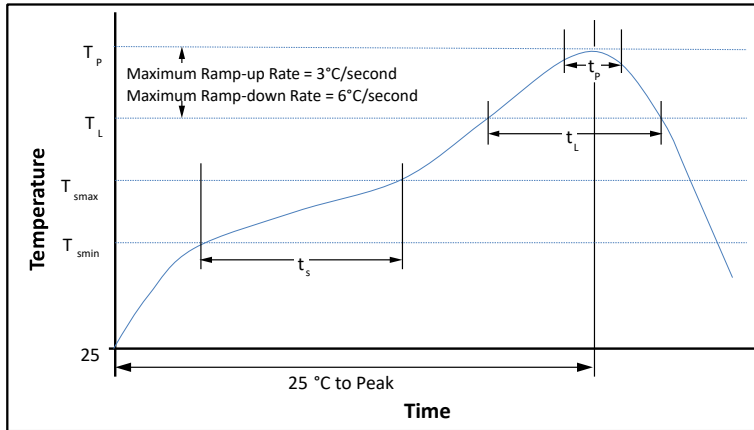
Catalog numbering system

Taping



(Note 1) Reflow Conditions

Reflow parameters and reflow profile are shown as following,

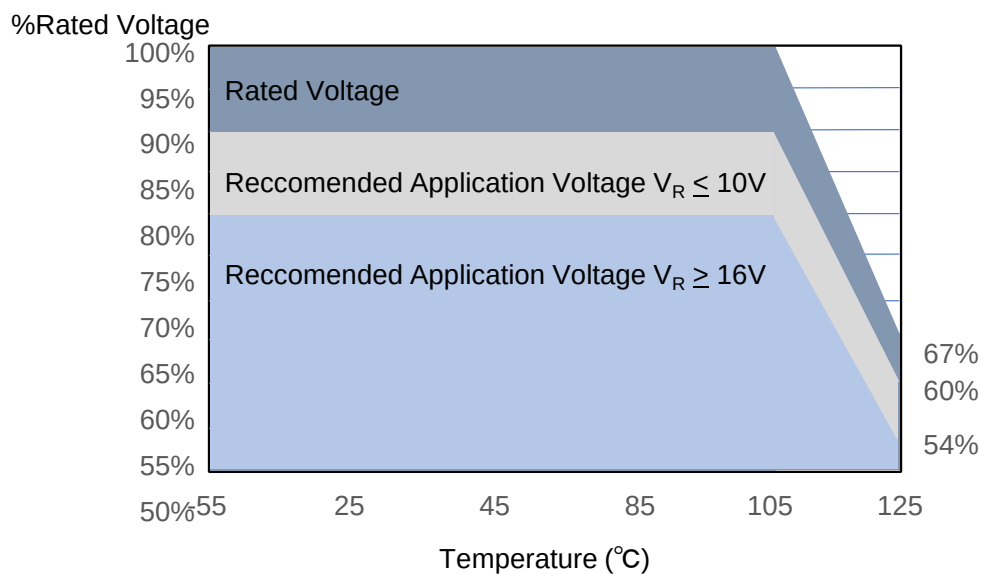


Profile Feature	Pb-Free Assembly
Preheat/Soak	
Temperature Minimum (T_{Smin})	150°C
Temperature Maximum (T_{Smax})	200°C
Time (t_s) from T_{Smin} to T_{Smax}	60 – 120 seconds
Ramp-up Rate (T_L to T_{Smax})	3°C/seconds maximum
Liquidous Temperature (T_L)	217°C
Time Above Liquidous (t_l)	60 – 150 seconds
Peak Temperature (T_P)	260°C
Time within 5°C of Maximum Peak Temperature (t_p)	30 seconds maximum
Ramp-down Rate (T_P to T_L)	6°C/seconds maximum
Time 25°C to Peak Temperature	8 minutes maximum

Number of reflow cycles: 2
MSL: 3

Note: All temperatures refer to the center of the package, measured on the package body surface that is facing up during assembly reflow.

(Note 2) Derating recommendation



(Note 3) Halogen-free, Lead-free, & RoHS Compliant

NO.	ITEM		PERFORMANCE	TEST CONDITION
1	Operating Temperature		-55 to +125deg.C	
2	Rated Voltage [V.dc] (=Category Voltage)		25	Temperature: 125deg.C
3	Surge Voltage [V.dc]		29.0	Temperature: 125deg.C
4	Capacitance		See standard ratings	Frequency: 120Hz Measurement volt.:0.5Vrms+1.5Vdc
5	Capacitance tolerance		-/+20 %	Frequency: 120Hz Measurement volt.:0.5Vrms+1.5Vdc
6	Dissipation Factor (tanD)		See standard ratings	Frequency: 120Hz Measurement volt.:0.5Vrms+1.5Vdc
7	Leakage Current (L.C.)		See standard ratings	Rated voltage for 5 min.
8	Equivalent series resistance (ESR)		See standard ratings	Frequency:100kHz
9	Dimensions		Mechanical dimensions shall meet the Requirements specified on drawing.	
10	Endurance	ΔC/C	Within -20%/+20% of initial value	125°C at 2/3 rated voltage, 2,000 hours
		DF	Within 1.5 × initial limit	
		LC	Within 2 × initial limit	
		ESR	Within 2 × initial limit	
11	Storage Life	ΔC/C	Within -20%/+20% of initial value	125°C at 0 volts, 2,000 hours
		DF	Within 1.5 × initial limit	
		LC	Within 2 × initial limit	
		ESR	Within 2 × initial limit	
12	Humidity	ΔC/C	Within -20%/+40% of initial value	60°C, 90~95% RH, 0V, 500 hours
		DF	Within 1.5 × initial limit	
		LC	Within 3 × initial limit	
		ESR	N/A	
13	Surge Voltage	ΔC/C	Within -20%/+20% of initial value	125°C, 1.32 × (0.67×rated voltage), 1000 cycles, 33 Ω in series
		DF	Within initial limit	
		LC	Within initial limit	
		ESR	Within initial limit	

NO.	ITEM	PERFORMANCE	TEST CONDITION
14	Temperature Stability	$\Delta C/C$	Within initial limit
		DF	Within initial limit
		LC	Within initial limit
		$\Delta C/C$	Within -20%/+20% of initial value
		DF	Within initial limit
		$\Delta C/C$	Within -20%/+50% of initial value
		DF	Within 1.2 × initial limit
		LC	Within 10 × initial limit
		$\Delta C/C$	Within -30%/+50% of initial value
		DF	Within 1.5 × initial limit
		LC	Within 10 × initial limit
15	Mechanical Shock / Vibration	$\Delta C/C$	Within -10%/+10% of initial value
		DF	Within initial limit
		LC	Within initial limit
16	Failure rate level	0.5%/1000hr (CL 60%)	Temperature: 105 +/-2 deg. Rated voltage applied. Series resistance: 3ohm

<In the case of doubt in measured value, please measure again after treatment as following>

DC protective resistance : 1000Ω,

Applied Voltage : Rated Voltage,

Temperature : 105C

Charging Time : 1 hour

Please measure after keeping 24-48hour in the room temperature and humidity after above treatment applying.